AZIZ ASHFAK

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ACADEMIC QUALIFICATIONS

University	Noakhali Science and Technology University, Noakhali, Chattogram, Bangladesh
Degree	B.Sc. (Honours) in Statistics | CGPA 3.37 / 4.00 | Apr 2022 – Aug 2026
Thesis	BanglaStatMCQ-500: A Statistical Reasoning Benchmark in Bangla for Open-Source Large Language Models
Supervisor	Md. Rasel Hossain, Assistant Professor, Department of Statistics
College Certificate	Narsingdi Govt. College, Narsingdi, Dhaka, Bangladesh Higher Secondary School Certificate (HSC) | GPA 5.00 / 5.00 | 2021
School Certificate	Adiabad Islamia High School and College, Narsingdi, Dhaka, Bangladesh Secondary School Certificate (SSC) | GPA 5.00 / 5.00 | 2018

PROFESSIONAL EXPERIENCE

Research Associate, University of South Alabama *Jul 2025 – Present (16 months)*

Supervisor: Mahi Washim

Principal Responsibilities:

- Built NLP pipelines for scientific document ingestion in Python and LangChain.
- Engineered retrieval-augmented generation (RAG) systems with OpenAI and Groq.
- Developed multimodal RAG with BLIP-2 and VQA models to interpret figures.
- Fine-tuned Hugging Face Transformers for claim extraction and classification.
- Applied Vision Transformers (ViT) in PyTorch to document-layout understanding.
- Architected a PostgreSQL store on Docker for entities, concepts and edges.
- Indexed corpora for semantic retrieval with FAISS and ChromaDB.
- Constructed evidence-weighted fuzzy cognitive maps (FCM) in Python.
- Evaluated RAG and FCM quality with LLM-as-judge against verified ground truth.

AI Engineer, ROYALX LLC, Dhaka, Bangladesh (remote internship) *Mar 2025 – Jun 2025 (4 months)*

Principal Responsibilities:

- Deployed production LLM applications on OpenAI, Groq and Hugging Face.
- Built multimodal RAG pipelines end to end with LangChain and LlamaIndex.
- Fine-tuned transformer models in PyTorch for question answering and tagging.
- Developed NLP modules for summarisation, extraction and semantic search.
- Architected LangChain and LangGraph agents for multi-step reasoning.
- Served models through ONNX Runtime for optimised inference.
- Optimised vector search with FAISS and ChromaDB for low-latency inference.
- Shipped services on FastAPI and Streamlit with GitHub Actions CI/CD.

Freelance Data Analyst, Hestia Traffic App Project *Dec 2024 – Feb 2025 (3 months)*

Principal Responsibilities:

- Analysed large-scale traffic and road-accident data in Python.
- Engineered cleaning and preprocessing pipelines with pandas and NumPy.
- Ran exploratory analysis on violations, frequency and road conditions.
- Visualised accident patterns with Matplotlib and Seaborn.
- Conducted statistical testing of variables influencing road safety in R.

- Translated findings into recommendations for a traffic-monitoring platform.

Research Assistant, Noakhali Science and Technology University *Sep 2023 – Nov 2024 (15 months)*

Focus: *Machine Learning, Deep Learning and Representation Learning*

Supervisor: Md. Rasel Hossain, Assistant Professor, Department of Statistics

Principal Responsibilities:

- Collected and cleaned primary survey data with pandas and NumPy.
- Ran exploratory data analysis on distributions, missingness and outliers.
- Visualised findings with Matplotlib and Seaborn.
- Built supervised learning pipelines with scikit-learn over tabular data.
- Benchmarked ensemble methods against linear baselines under matched inputs.
- Trained CNNs in PyTorch and TensorFlow for image classification.
- Fine-tuned ResNet, DenseNet and EfficientNet backbones for segmentation.
- Applied Vision Transformers and self-supervised pretraining to sparse data.

PUBLICATIONS

Accepted / Peer-Reviewed:

- "Optimizing U-Net Performance: Evaluation of State-of-the-Art Encoders and Optimizers on Human Segmentation Dataset," in Proc. 2026 IEEE Int. Conf. Systems, Man, and Cybernetics (SMC), Bellevue, WA, USA, Oct. 2026, to appear in IEEE Xplore. (Co-author.)

Manuscripts in process:

- **A. Ashfak**, K. A. Siyam, M. R. Hossain, R. Hossain, M. Washim, and I. J. Nikhat, "Robustness of CNN Transfer Learning versus Vision Transformers for Aerial Scene Classification," submitted to ICCIT 2026, under review.
- **A. Ashfak**, K. A. Siyam, M. R. Hossain, and M. Washim, "When Does Relevance-Aware Retrieval Help? An Analysis of Multimodal RAG for Scientific Visual Question Answering," manuscript in preparation, 2026.
- **A. Ashfak**, M. R. Hossain, and K. A. Siyam, "TopoVLM: Testing Whether Vision-Language Models Understand Phylogenetic Topology," manuscript in preparation, 2026.
- M. S. Hossain, M. R. Hossain, R. Hossain, **A. Ashfak**, I. Islam, and S. A. Ayon, "Skilled Birth Attendance and Height-for-Age Among Children Aged 0–23 Months in Bangladesh: A Survey-Weighted Targeted Maximum Likelihood Analysis of the 2025 Multiple Indicator Cluster Survey," submitted to BMC Public Health, under review, 2026.
- R. Hossain, M. S. Hossain, M. R. Hossain, S. A. Ayon, and **A. Ashfak**, "Functional Difficulty and ICT Skill Attainment Among Women in Bangladesh: Evidence from the 2025 Multiple Indicator Cluster Survey," submitted to BMC Public Health, under review, 2026.

SELECTED PROJECTS

Multimodal Research Paper Analyst [code]

- Built multimodal RAG over paper text and figures with Groq and Llama.
- Interpreted charts and tables with vision-language models.
- Shipped a Streamlit app exporting to Word, PDF, CSV and Markdown.

Brain Tumor Detection [code] [live]

- Benchmarked LeNet, ResNet152V2 and DenseNet121 to 98.52% accuracy in TensorFlow.
- Exported the winning backbone to ONNX for low-latency inference.
- Deployed on Render with Flask and GitHub Actions CI/CD.

Face Mask Detection [code]

- Trained a real-time detector on an 853-image set with YOLO in PyTorch.
- Localised multiple faces per frame with OpenCV preprocessing.
- Served through Flask with ONNX Runtime for optimised inference.

TECHNICAL SKILLS

Programming	Python, SQL, R, C (basic), Java (basic), LaTeX
Statistics	Hypothesis testing, regression, statistical inference, experimental design; bootstrap resampling, permutation tests, FDR correction; survey-weighted estimation, latent class analysis; exploratory data analysis
Machine learning	scikit-learn; ensemble methods; feature engineering, model selection, hyperparameter tuning, cross-validated evaluation
Deep learning	PyTorch, TensorFlow, Keras; CNNs (LeNet, ResNet, DenseNet, EfficientNet); U-Net segmentation; YOLO detection; Vision Transformers; transfer learning, self-supervised pretraining, PEFT/LoRA
LLMs and RAG	OpenAI API, Groq, Hugging Face Transformers, LangChain, LangGraph, LlamaIndex; vision-language models (BLIP-2, VQA); multimodal, agentic and knowledge-graph RAG; prompt engineering
Retrieval and databases	FAISS, ChromaDB, Pinecone; PostgreSQL, SQLite; semantic and vector retrieval
Deployment and MLOps	Docker, FastAPI, Flask, Streamlit, Gradio; ONNX export; Git and GitHub Actions CI/CD; Render; ETL pipeline design
Data and visualisation	pandas, NumPy, Matplotlib, Seaborn, OpenCV; Jupyter, Google Colab, Kaggle

ACHIEVEMENTS & CO-CURRICULAR ACTIVITIES

Positions Held:

- Research Instructor, NSTU Research Society (January 2024 – August 2025)
- General Secretary, NSTU Data Science Club (February 2026 – August 2026)
- Treasurer, NSTU Blood Brigade Society (January 2023 – November 2023)

Campus Activities:

- Organised and delivered departmental workshops on machine learning and statistical computing for undergraduate students.
- Mentored junior students through their first research projects and conference submissions.
- **Organiser, National Statistics Day Celebrations,** Noakhali Science and Technology University | 2025, 2026 — contributed to organising university events promoting statistical awareness and education.
- Attended the 3rd International Conference on Machine Intelligence and Emerging Technologies (MIET 2026), NSTU, Bangladesh.
- **Volunteer, NSTU Research Fair 2025** — supported the organisation and execution of a university research fair.

Memberships:

- Member, NSTU Statistics Society
- Member, NSTU Blood Brigade Society

Awards & Achievements:

- **Kaggle —** published notebook with 165 upvotes and 348 forks, showing community engagement and reuse of shared data science work.
- **Runner-up, Statistics Premier League** (short-pitch cricket) — team captain; named Man of the Match in the semi-final.
- **Junior School Certificate (JSC) General Grade Scholarship** — awarded for academic achievement.